

Xiaoyu Fan

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Education

2022 – present

New York University Stern School of Business, New York, NY

Ph.D. Candidate in Operations Management

Advisor: Zhengyuan Zhou

2018 – 2022

Nanjing University Mathematics Department, Nanjing, Jiangsu, China

B.S. in Information and Computational Science

Publications

1. Sample Complexity of Policy Learning for Inventory Control with Censored Demand

Xiaoyu Fan, Boxiao Chen and Zhengyuan Zhou

Second Round Major Revision at Management Science

** Finalist, INFORMS Undergraduate Operations Research Prize, 2022*

2. No-Regret Learning in Multi-Retailer Inventory Control

Xiaoyu Fan, Boxiao Chen, Wenqiang Xiao and Zhengyuan Zhou

Third Round Focused Major Revision at Management Science

3. Sample Complexity of Inventory Control with Setup Costs Beyond Plug-in Demand Estimation

Xiaoyu Fan and Zhengyuan Zhou

Major Revision at Manufacturing & Service Operations Management

** Finalist, Data Mining Best Student Paper Competition, 2025*

** Finalist, INFORMS JFIG Paper Competition, 2024*

4. Learning to Bid in Non-Stationary Repeated First-Price Auctions

Zihao Hu, Xiaoyu Fan, Yuan Yao, Jiheng Zhang and Zhengyuan Zhou

Minor Revision at Operations Research

5. Product Return as a Sequence of Search Processes: Optimality and Search Duration

Xiaoyu Fan, Srikanth Jagabathula, Zijian Liu and Eitan Muller

Second Round Major Revision at Operations Research

6. **Personalized Dynamic Pricing with Demand Learning: The Impact of Growing Effective Dimensionality**
 Gah-Yi Ban, Xiaoyu Fan, Yanjun Han, Srikanth Jagabathula, N. Bora Keskin and Zhengyuan Zhou
Major Revision at Operations Research
7. **Don't Follow RL Blindly: Lower Sample Complexity of Learning Optimal Inventory Control Policies with Fixed Ordering Costs**
 Xiaoyu Fan, Boxiao Chen, Tava Lennon Olsen, Hanzhang Qin and Zhengyuan Zhou
Production and Operations Management, 2025

Working Papers

1. **Towards Optimal Inventory Control under Censored Demand: A Biased Sample-Average Approximation Approach**
 Yuxuan Han, Xiaoyu Fan, Jiawei Zhang and Zhengyuan Zhou
Under Review at Operations Research
2. **Optimal Sample Complexity of Distributionally Robust Direct Preference Optimization**
 Xiaoyu Fan, Will Ma, Sean X. Zhou and Zhengyuan Zhou
3. **Optimal Sample Complexity of Fine-Tuning LLM's Last Layer via Direct Preference Optimization**
 Xiaoyu Fan, Sean X. Zhou and Zhengyuan Zhou
4. **Leveraging Product Variations for Personalized Dynamic Pricing**
 Yanjun Han, Xiaoyu Fan, Srikanth Jagabathula and Zhengyuan Zhou
5. **Generative AI, Model Collapse and Intellectual Property Policy**
 Xiaoyu Fan, Arun Sundararajan and Zhengyuan Zhou

Awards and Honors

- Winner, AI Dissertation Proposal Competition, Texas A&M University, 2026
- NYU Stern CGEB PhD Grant, 2026
- Finalist, Data Mining Best Student Paper Competition, 2025
- Finalist, INFORMS JFIG Paper Competition, 2024
- Finalist, INFORMS Undergraduate Operations Research Prize, 2022

Teaching

- **OPMG-GB3321: Stochastic Processes** (Spring 2026) — Teaching Fellow

- **MULT-UB7: Decision Models and Analytics** (Summer 2025) — Instructor
 NYU Stern undergraduate core
 Teaching evaluation: 4.88/5.00 Class size: 15 Response rate: 53%
 Student comments:
 - *“The course content is very good. The instructor’s lectures were very clear.”*
 - *“Extremely helpful and open during office hours. Takes her time on students during office hours to make sure we understand the concept. Helped make the class a little less intimidating to being very doable.”*
 - *“Professor Fan, thank you very much for your instruction and ability to help us understand challenging topics.”*
- **OPMG-GB2350: Decision Models and Analytics** (Fall 2024) — Teaching Fellow
- **SHBI-GB7301: Stochastic Modeling & Simulation** (Summer 2024) — Teaching Fellow
- **COR1-GB2114: Operations** (Spring 2024) — Teaching Fellow

Seminars and Invited Talks

- AI Dissertation Proposal Competition, Texas A&M University, Sep 2026
- MOILS Seminar, NYU Stern, Apr 2026
- POMS-HK International Conference, Jan 2026
- INFORMS Annual Meeting, Oct 2025
- NYC Ops Day PhD Colloquium, Mar 2025
- INFORMS Annual Meeting, Oct 2024
- MOILS Seminar, NYU Stern, Sep 2024
- ICSA China Conference, Jun 2024
- POMS-HK International Conference, Jan 2024
- INFORMS Annual Meeting, Oct 2023
- JSM Conference, Aug 2023
- Nanjing University, Jul 2023
- POMS Conference, May 2023
- MOILS Seminar, NYU Stern, Apr 2023
- INFORMS Annual Meeting, Oct 2022

Professional Services

- Organizer:
 - AI Cities Workshop, TAU&NYU, 2026
 - MOILS Seminar, NYU Stern, 2024
- Reviewer: Management Science, Manufacturing & Service Operations Management, Production and Operations Management, Journal of Machine Learning Research, Annals of Operations Research, International Conference on Learning Representations, SIAM Journal on Optimization, IEEE Open Journal of Control Systems, IEEE Robotics and Automation Letters, Operations Research Letters, Conference on Decision and Control